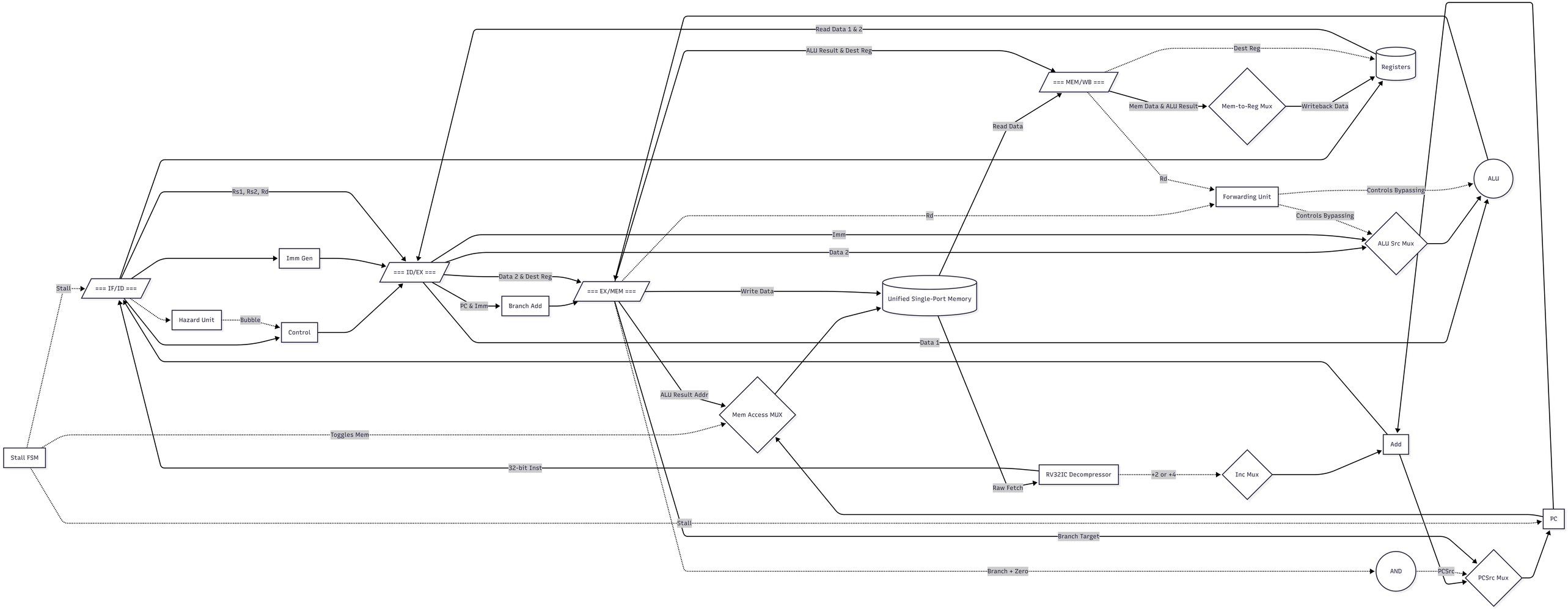
Read Data 1 & 2
ALU Result & Dest Reg
Dest Reg
Registers
=== MEM/WB ===
Mem Data & ALU Result
Mem-to-Reg Mux
Writeback Data
Read Data
Rd
Forwarding Unit
Controls Bypassing
Controls Bypassing
ALU
Rs1, Rs2, Rd
Rd
Imm
Data 2
ALU Src Mux
Imm Gen
=== ID/EX ===
Data 2 & Dest Reg
=== EX/MEM ===
Write Data
Unified Single-Port Memory
Stall
=== IF/ID ===
PC & Imm
Branch Add
Hazard Unit
Bubble
Control
Data 1
ALU Result Addr
Mem Access MUX
Toggles Mem
Add
Stall FSM
32-bit Inst
RV32IC Decompressor
+2 or +4
Inc Mux
Raw Fetch
Stall
PC
Branch Target
Branch + Zero
AND
PCSrc
PCSrc Mux